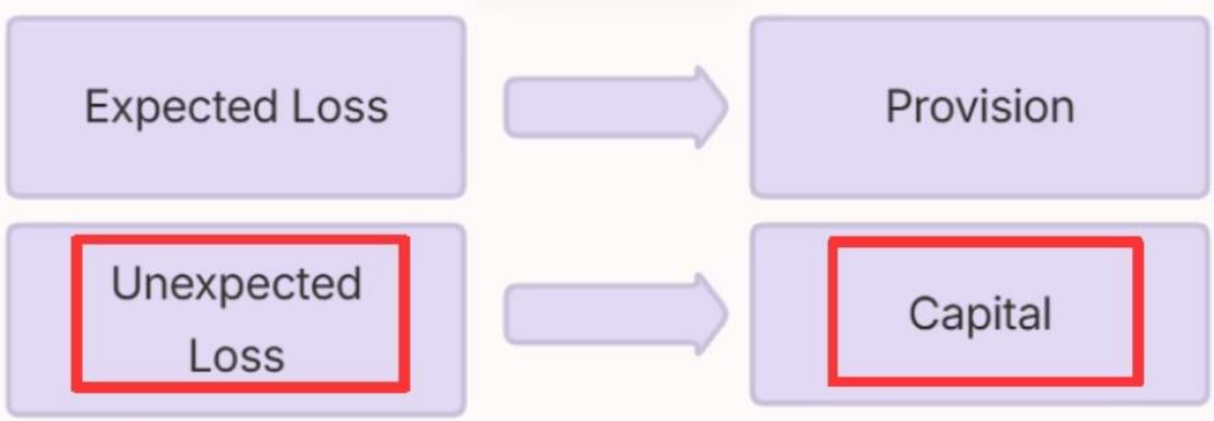
Application of AI for Basel Credit Risk Modelling in Banking Industry with HKMA Regulatory Compliance Analysis

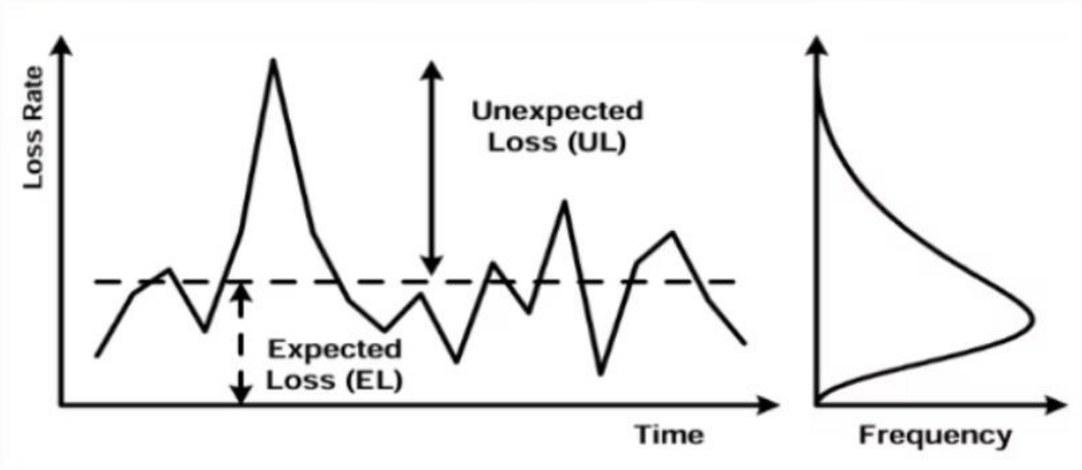
Agenda

- Background
- User Journey Map of Basel Modelling
- AI Applications on Mode Build
- AI Applications on Data, Reporting and Compliance
- Conclusion and Reflection

Background

Basel Modelling





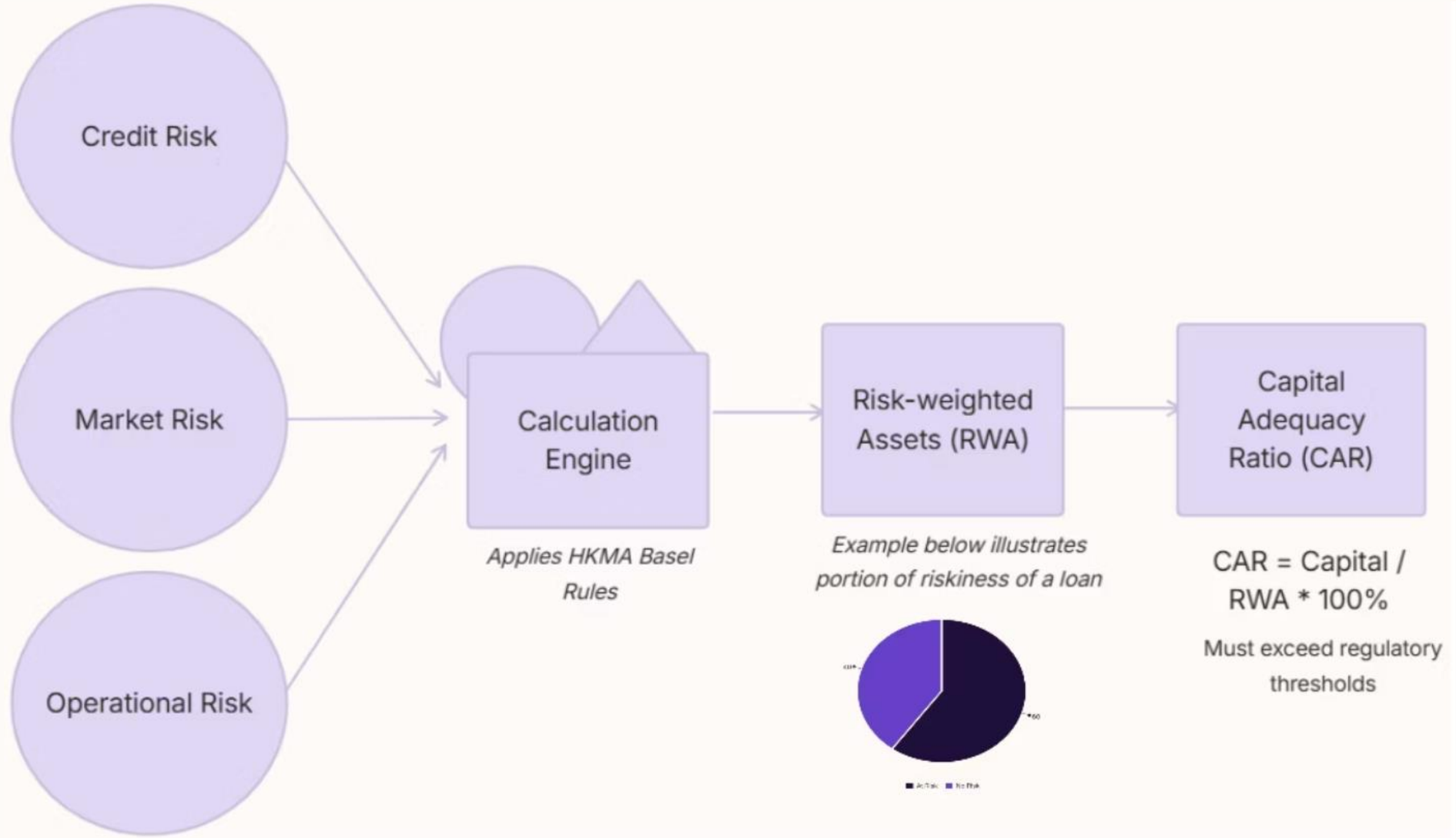
Basel Framework

- Global Standards ensure banks maintain adequate capital to absorb unexpected losses during financial stress (e.g. 2008 financial tsunami)

HKMA Banking Capital Rules

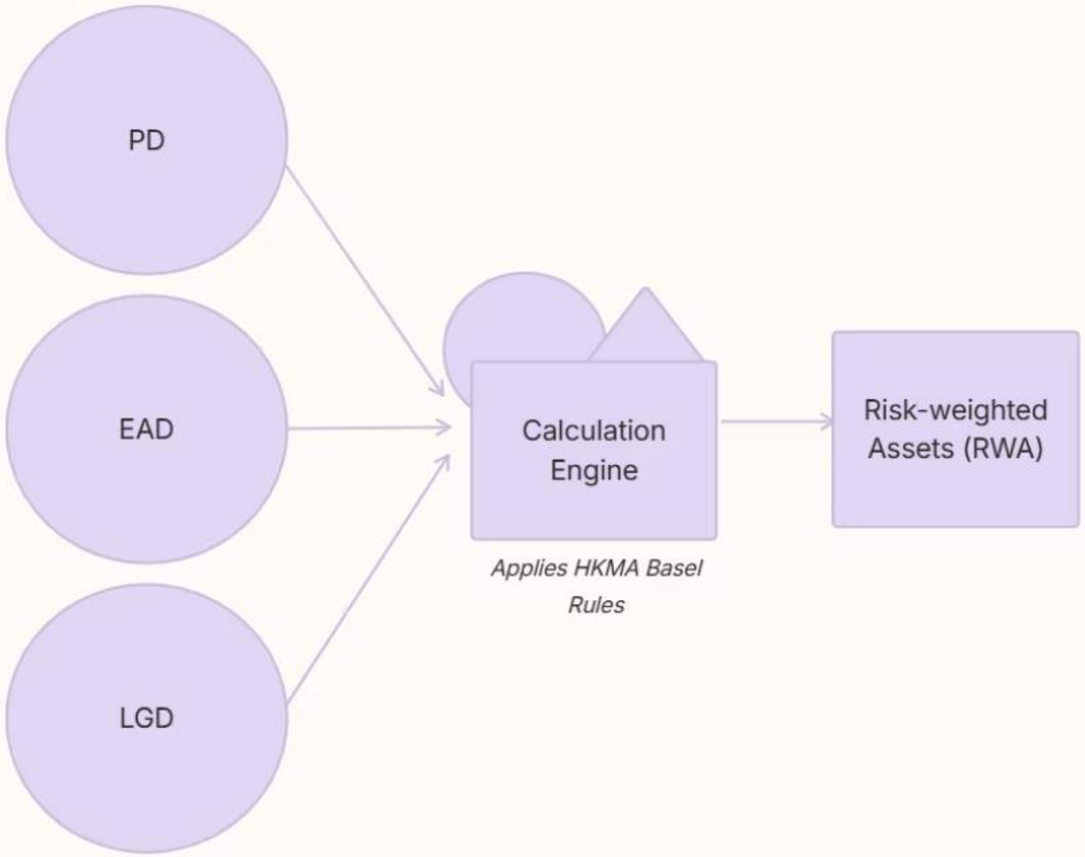
- Local adaption of Basel Framework by HKMA
- Tailored to address Hong Kong's banking risks and regulatory environment

Basel Modelling



Basel Modelling

Under **Credit Risk**, if using IRB Approach, there are 3 components input to engine:



Internal Ratings-Based (IRB) Approach

Bank needs to estimate:

- **PD (Probability of Default):** Likelihood of borrower default.
- **LGD (Loss Given Default):** Loss severity post-default.
- **EAD (Exposure at Default):** Total exposure at default.

for RWA calculations.

Standardized Approach (SA)

Rule-based, uses external ratings (simpler but less risk-sensitive).

Regtech Adoption Practice Guide

Issue #6: Artificial Intelligence-based Regtech Solutions

April 2022

HONG KONG MONETARY AUTHORITY
香港金融管理局

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Regtech Adoption Practice Guide is a publication published by the Hong Kong Monetary Authority (HKMA). It should be noted that the sole purpose of this publication is to provide Authorized Institutions (banks) with information on the latest regulatory technology (Regtech) developments. The HKMA does not endorse any use cases, solutions and/or implementation guidance described in this adoption practice guide. If a bank intends to adopt a particular solution or implementation, it should undertake its own due diligence to ensure that the technology or approach is suitable for its circumstances.

HKMA's view on AI in Financial Industry

HKMA initiatives:

- Nov 2020, HKMA released White Paper: *"Transforming Risk Management and Compliance: Harnessing the Power of RegTech"*.
- **Goal:** Roadmap to accelerate adoption of *Regulatory Technology (Regtech)* in Hong Kong.

What is Regtech?

- Utilizes innovative technologies to streamline compliance, regulatory reporting, and risk management.

AI in RegTech:

- *HKMA* released ***"Regtech Adoption Practice Guide - Issue #6: Artificial Intelligence-based Regtech Solutions"***
- It highlights how AI provides analytical power and automation capabilities that many Regtech solutions reply on.

Reference: Rechtech Adoption Practice Guide Issue #6 https://brdr.hkma.gov.hk/eng/doc-ldg/docId/getPdf/20220428-2-EN/20220428-2-EN.pdf

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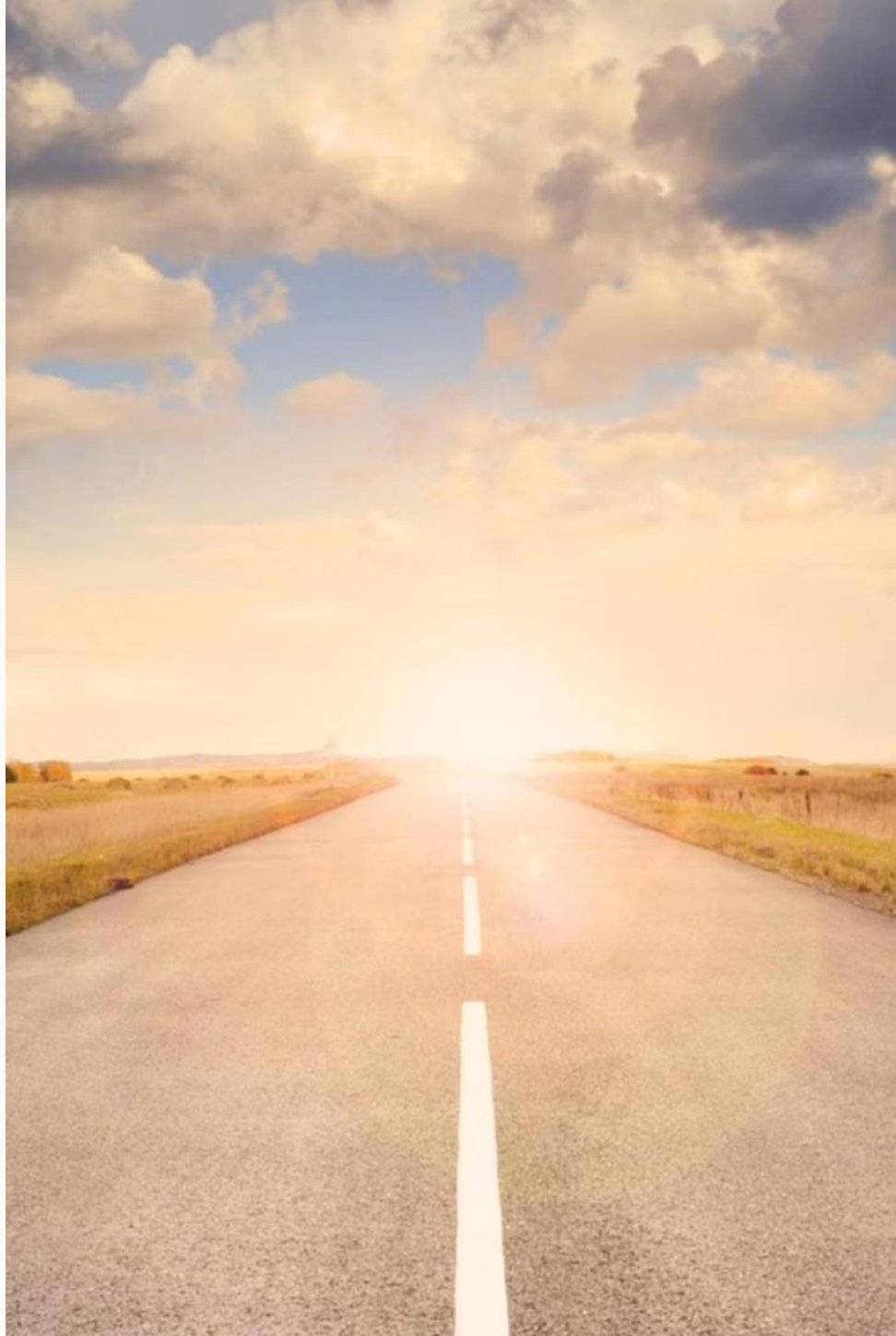
HKMA's view on AI in Financial Industry (cont'd)

Key risks:

- **Data Privacy and right of use**
  - Need privacy risk assessment, referencing PDPO
- **Accuracy and reliability**
  - Continuous monitoring if any model drift after deploy AI models
- **Explainability**
  - Implement adequate measures (e.g. SHAP) to ensure model transparency

User Journey Map for Basel Modelling



User Journey Map for Basel Modelling

Data Preparation

Backbone of reliable modelling

Model Building

Estimate PD, LGD, and EAD estimates using both traditional and advanced methods

Documentation

Model papers, CAR impact analyses, and respond to HKMA queries etc.

Compliance Reporting

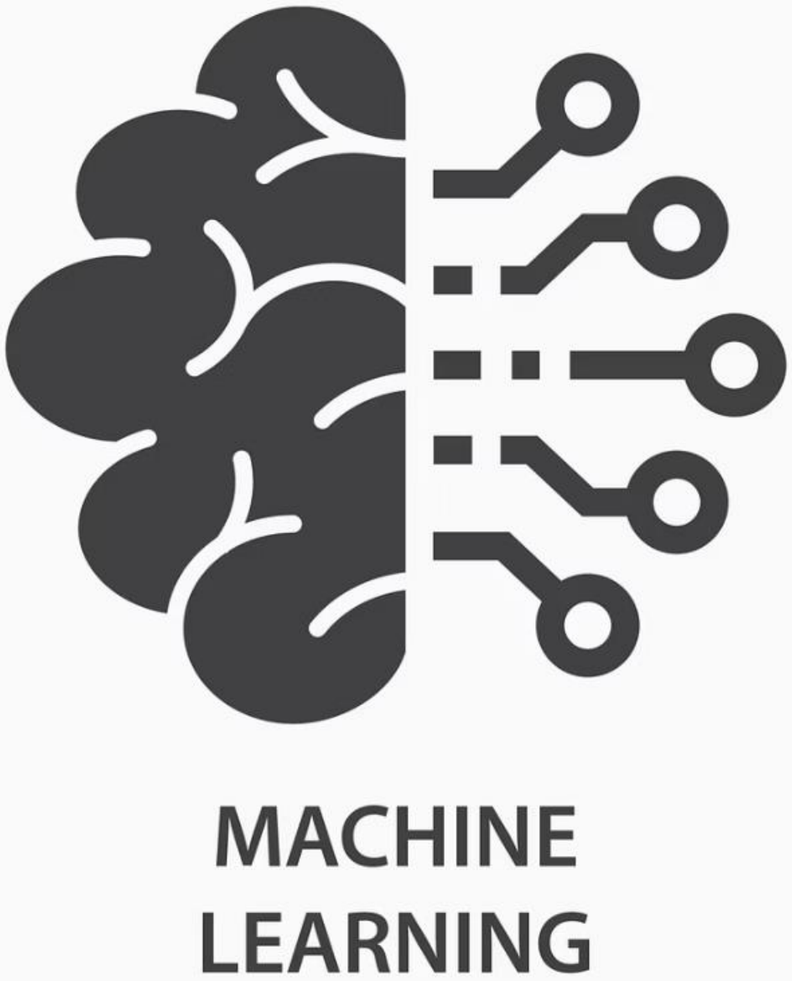
Pillar 3 disclosures, HKMA Survey etc.

Monitoring

Model performance, regulatory updates

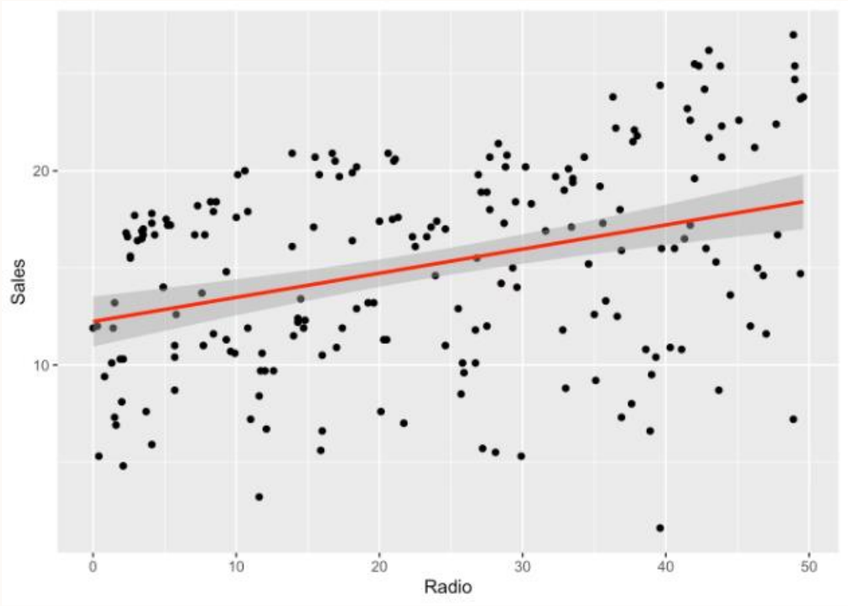
Feedback Loop: Continuous refinement based on HKMA guidelines and model drift analysis.

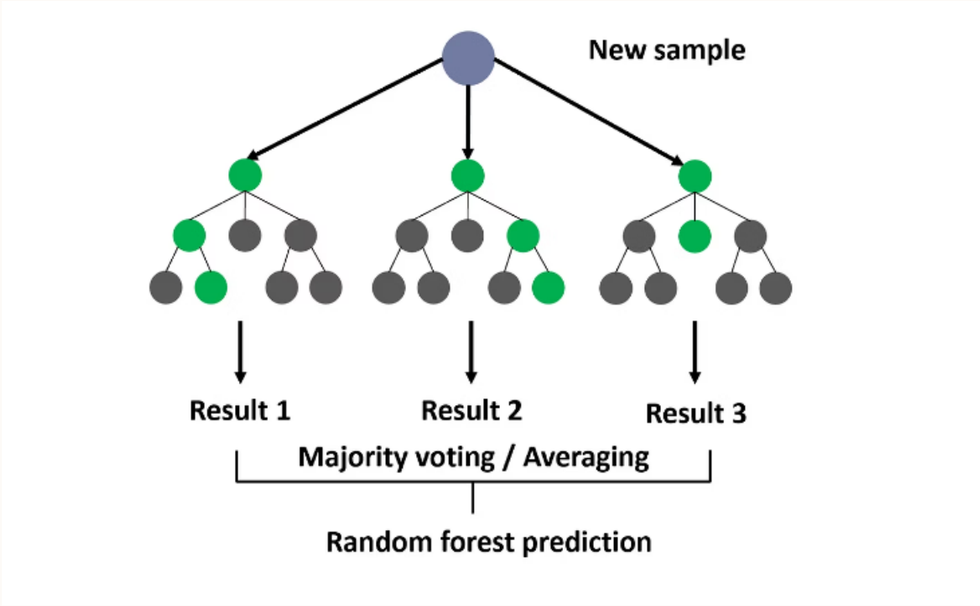
AI Applications on Model Build (Python Project)



Explore Advanced Machine Learning Algorithm

Focus: Compare traditional statistical models and advanced machine learning algorithms for PD prediction





Traditional Statistical Methods:

- Logistic Regression / Decision Tree
- Commonly used in PD model

Advanced Machine Learning Algorithms:

- Random Forest / XGBoost
- Rationale: Supervised Learning + Structured Data

Measurement: Use ROC Measure (AUC) to compare due to the PD is always with unbalanced data (defaults much less than non-defaults) and its robutness

Credit Risk Data Attributes (source from Kaggle)

Target Variable

- **Loan_Status:** Binary indicator (1 default vs. 0 non-default).

Features

- **Person_Age:** Age of the borrower.
- **Person_Income:** Annual income.
- **Person_Home_Ownership:** Categories include RENT, OWN, MORTGAGE, OTHER.
- **Person_Emp_Length:** Years of employment.
- **Loan_Intent:** Purpose of the loan (e.g., EDUCATION, MEDICAL, HOMEIMPROVEMENT, etc.).

- **Loan_Amnt:** Requested loan amount.
- **Loan_Int_Rate:** Interest rate on the loan.
- **Loan_Percent_Income:** Ratio depicting the requested loan amount relative to income.
- **CB_Person_Default_on_File:** Historical marker (Yes/No) for previous defaults.
- **CB_Person_Cred_Hist_Length:** Length in years of credit history.

Derived Features (in Feature Engineering)

- **Credit_Age_Ratio:** This ratio measures the proportion of a person's entire life during which they have had a credit history.
- **Employment Stability:** This score measures the ratio of the time a person has been employed to their available working years.

Have a glance on the data

	person_age	person_income	person_home_ownership	person_emp_length	loan_intent	loan_amnt
0	22	59000	RENT	123.0	PERSONAL	35000
1	21	9600	OWN	5.0	EDUCATION	1000
2	25	9600	MORTGAGE	1.0	MEDICAL	5500
3	23	65500	RENT	4.0	MEDICAL	35000
4	24	54400	RENT	8.0	MEDICAL	35000

loan_int_rate	loan_status	loan_percent_income	cb_person_default_on_file	cb_person_cred_hist_length
16.02	1	0.59	Y	3
11.14	0	0.10	N	2
12.87	1	0.57	N	3
15.23	1	0.53	N	2
14.27	1	0.55	Y	4

Reference: Kaggle Dataset https://www.kaggle.com/datasets/laotse/credit-risk-dataset

Model Life Cycle

With demonstrate on how the AI-driven Basel modelling algins with **SPM CA-G-4 'Validating Risk Rating Systems under the IRB Approach'**

1 Exploratory Data Analysis

- Understand data distributions, detect anomalies, and assess missing values.
- **Regulatory Compliance:**
  - **CA-G-4 Section 6.7:** Mandates data quality assessment before model development.

2 Data Cleansing

- Remove Duplicates / Missing Values / Outliers
- **Regulatory Compliance:**
  - **CA-G-4 Section 6.5:** Ensure that data are validated and cleansed.

3 Feature Engineering

- One-Hot Encoding / Standardization / Create New Features
- **Regulatory Compliance:**
  - **CA-G-4 Section 3.1:** Ensures logic and conceptual soundness of rating systems.

4 Model Training

- **Regulatory Compliance:**
  - **CA-G-4 Section 3.5:** Requires benchmarking models against industry standards.
  - **Experience with HKMA:** Justify your choice of model methodology / Multiple candidate models.

5 Model Testing

- Use ROC measure (AUC) to do validation
- **Regulatory Compliance:**
  - **CA-G-4 Section 7.2:** Mandate to validate discriminatory power of PD model. It also suggests AUC as one of the measurement

6 Model Explainability

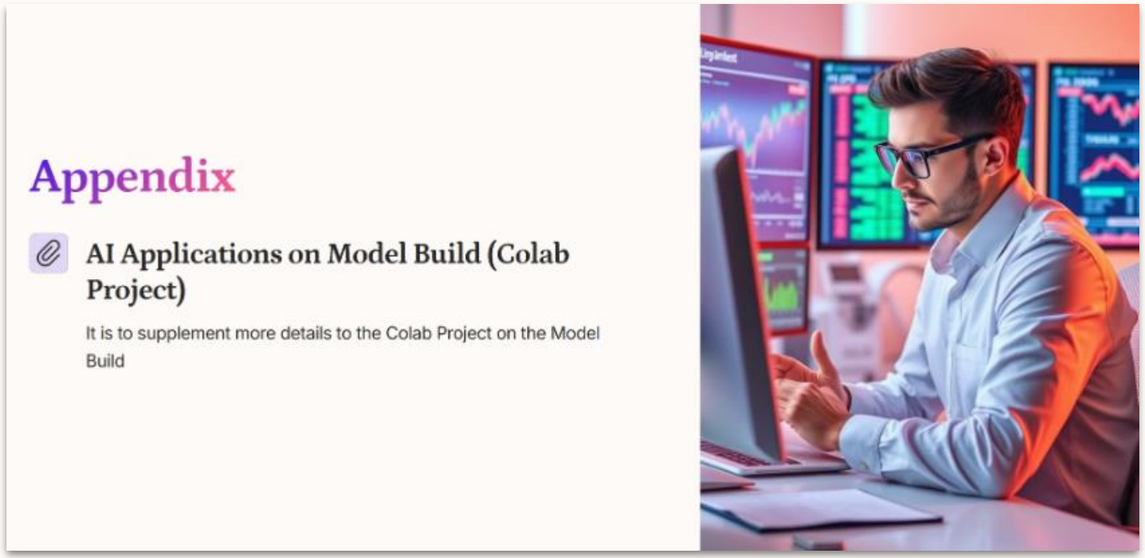
- Use SHAP for model explainability
- **Regulatory Compliance:**
  - **CA-G-4 Section 5.2:** Requires transparency in rating system.

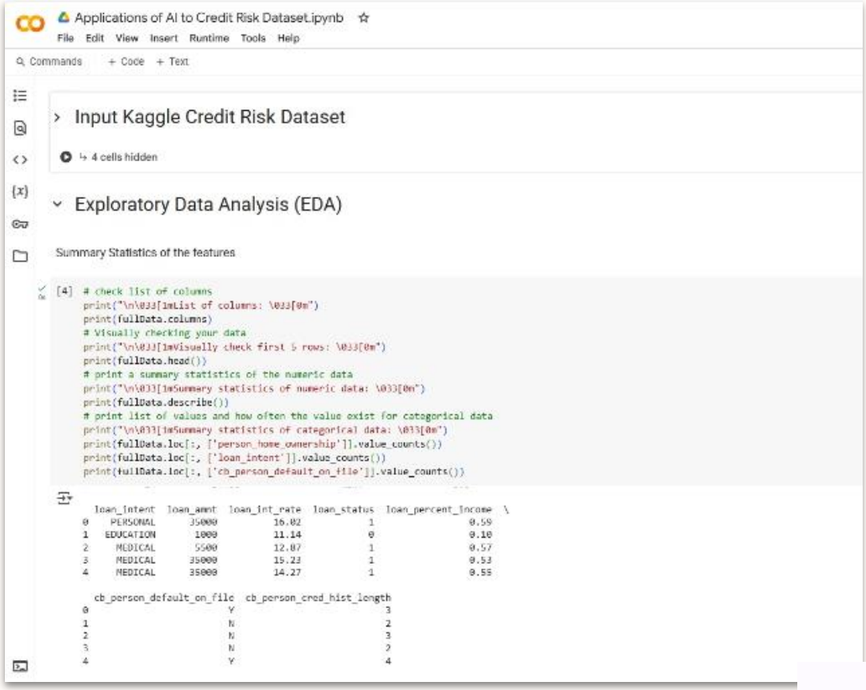
Note: *SPM CA-G-4 is a regulatory guideline issued by the* ***HKMA*** *that sets out the requirements for banks using* ***internal rating systems*** *to assess credit risk under the* ***Internal Ratings-Based (IRB) approach****, ensuring models are accurate, transparent, and aligned with international capital adequacy standards.*

https://brdr.hkma.gov.hk/eng/doc-ldg/docId/getPdf/20180517-2-EN/CA-G-4.pdf

More details of the Python Project

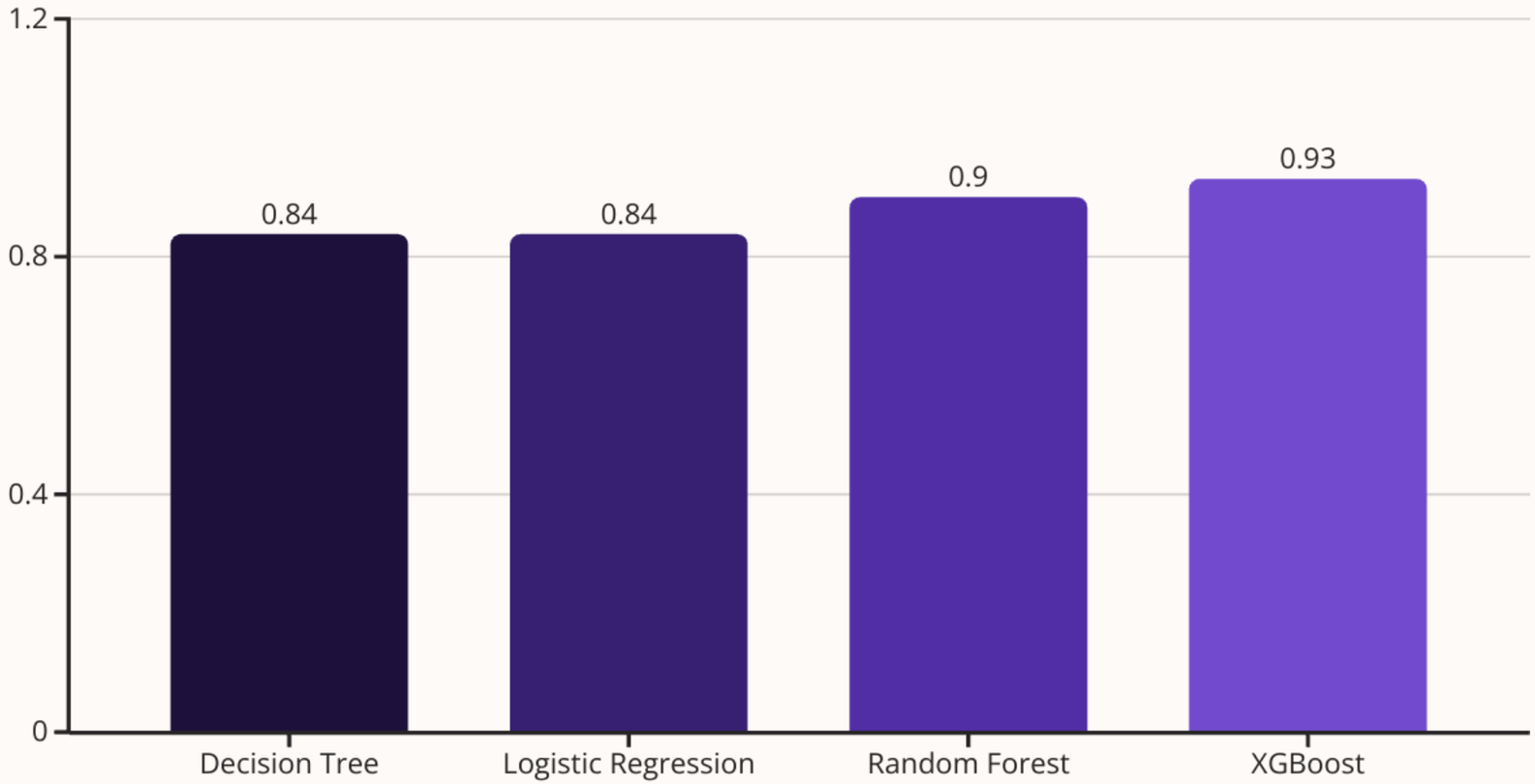
- As time is limited, for more details on Notebook like steps of data cleansing, feature engineering etc. of the Python Project, please refer to another power point slides *“Appendix”* and Python code *“credit_risk_modelling.ipynb”*
- In the coming slides, only the model results will be gone through





Model Performance Comparison

AUC Results (use 30% of data for testing)



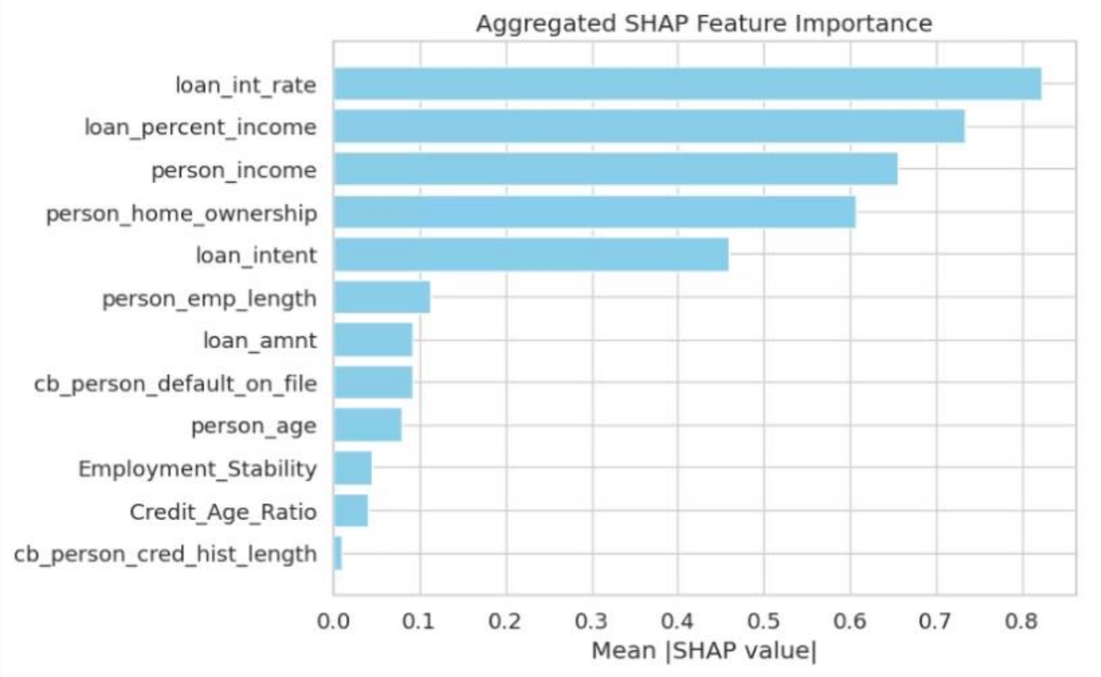
Insight: XGBoost with highest AUC score due to its boosted ensemble approach, offers superior performance.

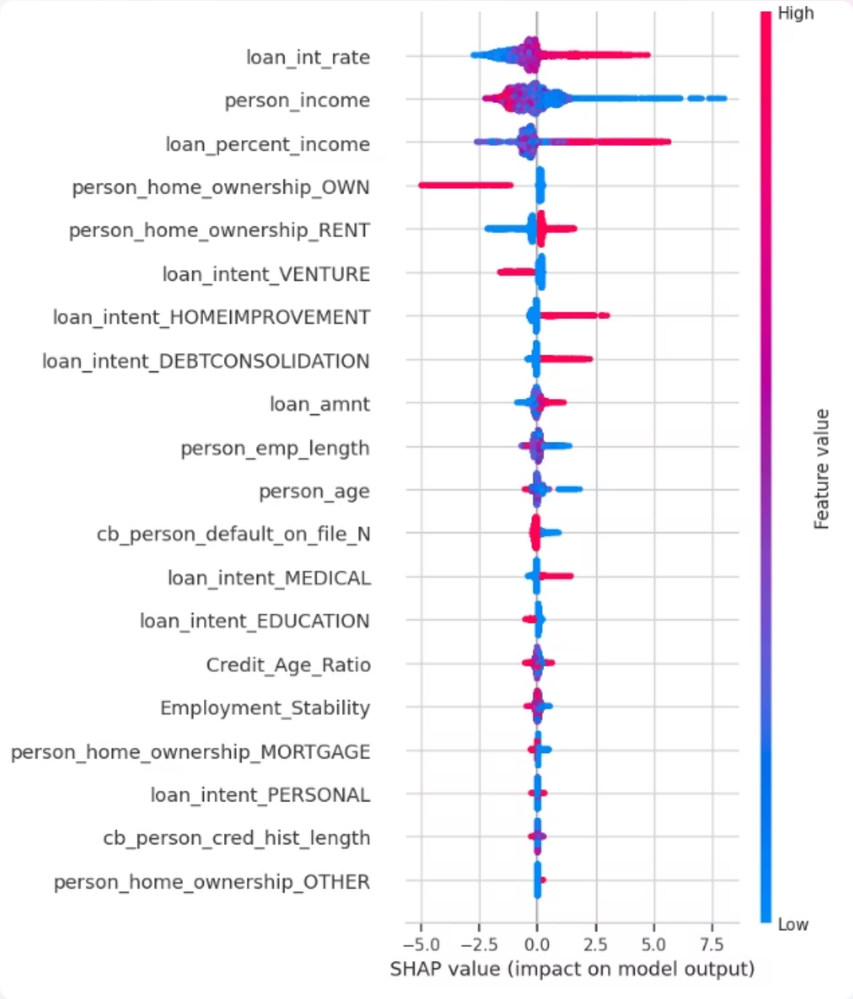
Model Explainability

We use SHAP to fulfill the HKMA's requirement on model transparency

SHAP Feature Importance

- Choose XGBoost model with highest AUC score to move forward to model explainability (SHAP)
- There are character features "person_home_ownership", "loan_intent", "cb_person_default_on_file" which have been applied One Hot Encoding.
- SHAP values in summary plot for One Hot Encoded features are grouped by averaging SHP values to easy investigate the feature importance
- **Observation:**
  - **loan_int_rate, loan_percent_income, person_income, person_home_ownership** and **loan_intent** are features which contribute the most in the model prediction





SHAP Summary Plot

Interpretation on the most important features:

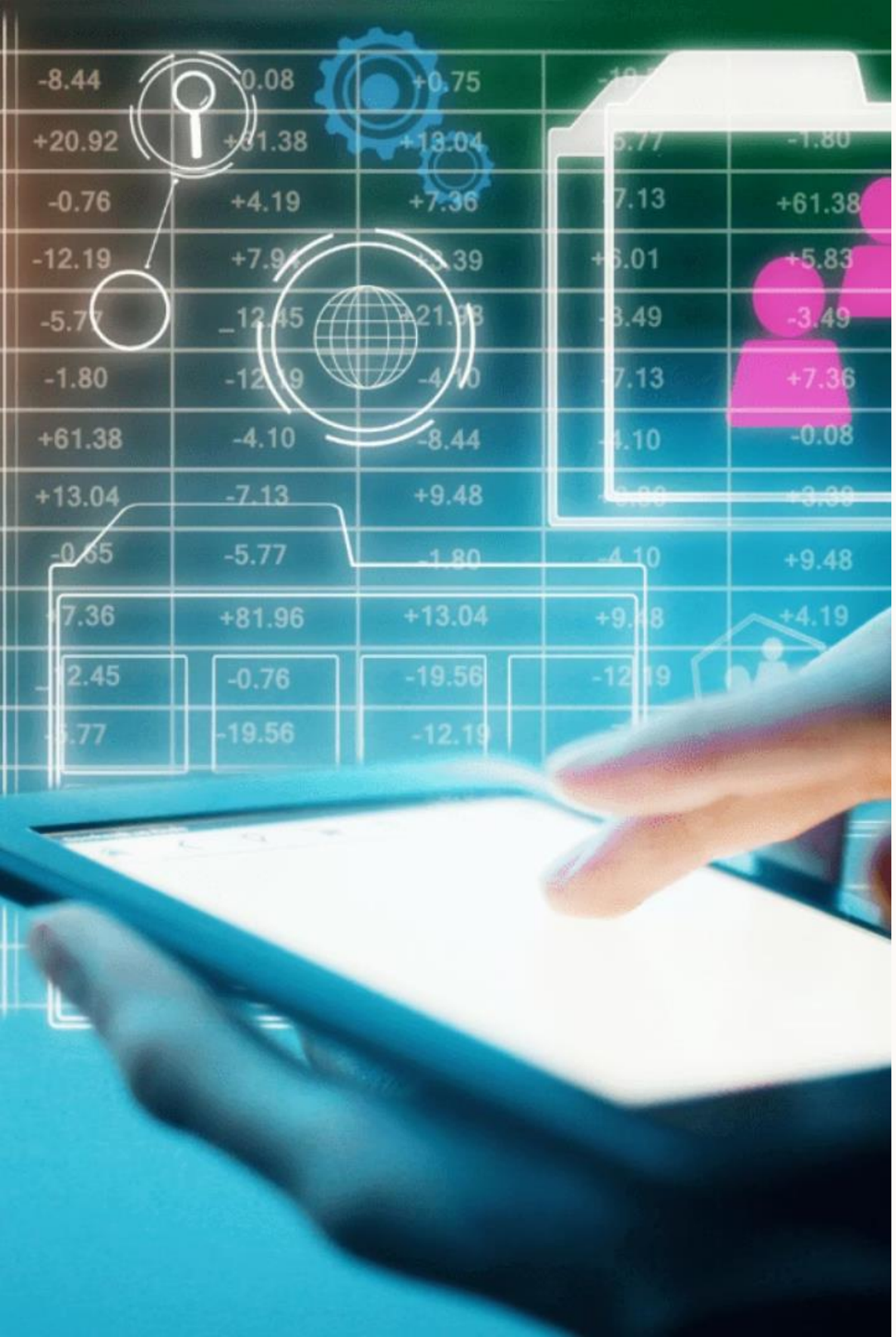
loan_int_rate: Borrower with higher interest rate with higher risk to default

loan_percent_income: Borrower with higher debt-to-income with higher risk to default

person_income: Borrower with higher income provides borrowers more means to cover their expenses and repay loans, hence lowering the risk

person_home_ownership: If a borrower owns their home, there's a strong stabilizing effect on the risk prediction. While renting tends to push the prediction toward higher risk. For other Mortgage and Other, these categories don't seem to contribute meaningful information.

loan_intent: If a borrower is seeking a Venture loan, it is associated with a lower predicted risk. Conversely, if a borrower is seeking Medical loan, it pushes to a higher risk. For other loan intents, they don't seem to contribute meaningful information.



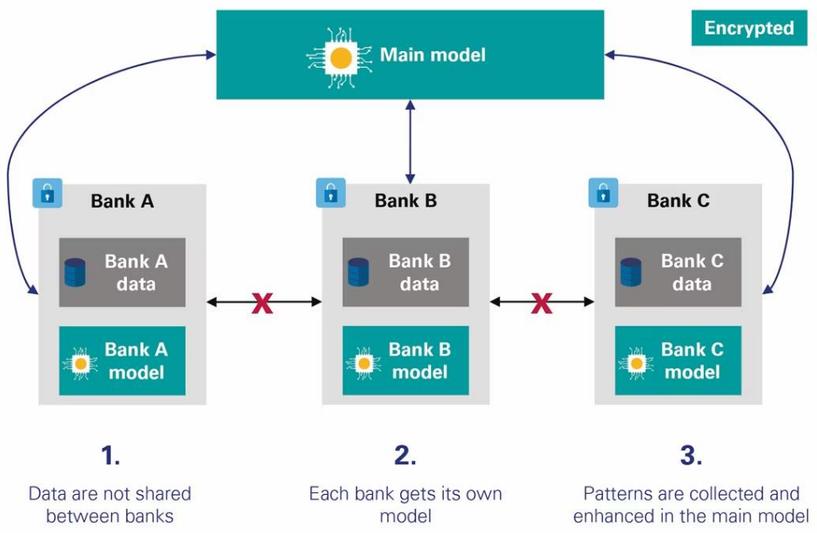
AI Applications on Data, Reporting and Compliance

Federated Learning

- From use case in **Regtech Adoption Practice Guide (Issue #6)**, Federated Learning technology allows participating banks to collaborate and build a machine learning model for **bankruptcy detection** together without disclosing their own data to each other

- The coming slides will introduce the pain point in data process in the Basel Modelling and how Federated Learning helps

Figure 4: Illustration of the Federated Learning network platform



Data Challenges under PIPL

Mainland China's Personal Information Protective Law (PIPL)

Prohibits cross-border data transfers without consent/certification

Financial Data Storage

Mandates data localization for critical sectors, including finance

Impact on Basel Models

Data fragmentation and impact on model accuracy

Reference: PIPL http://en.npc.gov.cn.cdurl.cn/2021-12/29/c_694559.htm

Federated Learning Solution for Compliance to PIPL

Train models locally on Mainland China data

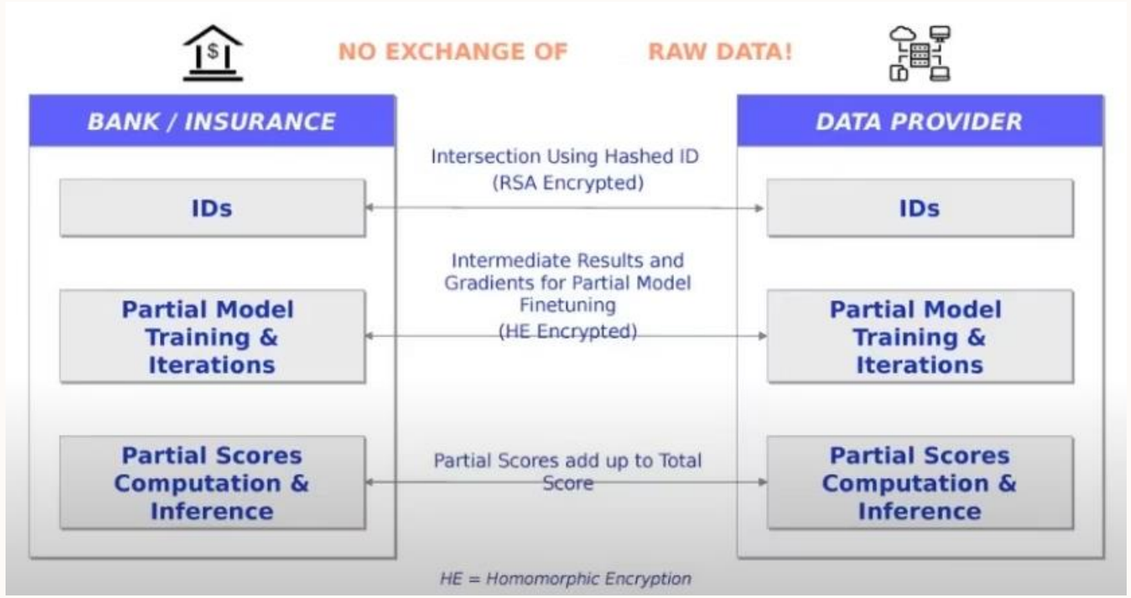
Models are trained locally without moving raw data across borders

Share encrypted model updates (not raw data)

Only model parameters and weights are transferred, maintaining data privacy

Aggregate updates into a global model

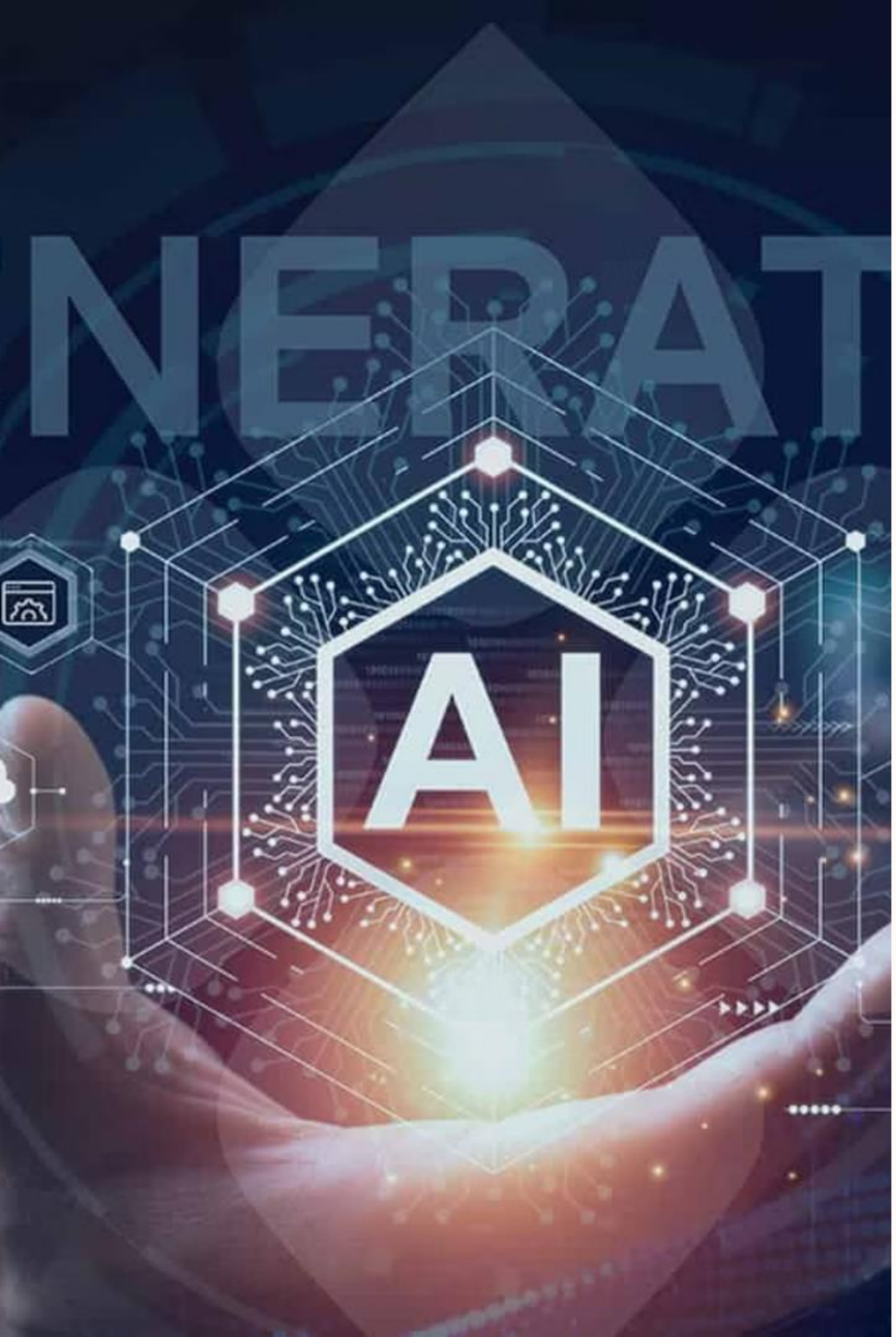
Combined model benefits from all data sources while complying with regulations



Benefits

- Meets the stringent requirements of PIPL.
- Aggregates insights from multiple data sources, improving model accuracy without compromising privacy.

Reference: Federated Learning https://www.youtube.com/watch?v=TpLTyyJmBc8&t=1002s



Generative AI

- In the coming slides, pain points in the Documentation & Compliance Analysis in Basel Modelling and how Generative AI solves them will also be discussed.
- HKMA's issued paper in 2024 on **“Generative Artificial Intelligence in the Financial Services Space”** will be discussed which provides a building block to help banks and other financial institutions to sail through the A.I. journey by sharing insights and practical experiences.

Pain Points in Documentation & Compliance Analysis

Documentation

Time-intensive for data retrieval on dozens of HKMA submissions.

Compliance Reporting

Frequent updates are required with significant manual overhead.

Monitoring & Feedback

Slow response to model drift and regulatory changes.

Impact to the team:

40%

Compliance Time

Spent on repetitive tasks

5-10%

Error Rate

In manual reports

AI-Driven Compliance Solutions

Smart Assistant

- Automatically generate regulatory and submissions checklists for model build/refinements.
- Analyze HKMA Q&A with knowledge base of HKMA requirements and internal model & data information

Automated Reporting

- Use GenAI-driven templates to auto-generate HKMA-compliant reporting.

Regulatory Feedback Loop Alert:

- Provide real-time alerts on new regulatory rules and suggested impact areas
- Monitoring thresholds are breached and suggested remedial actions

Benefits

- **Time Efficiency:**
  - Reduce documentation and report preparation from weeks to hours.
- **Error Reduction:**
  - Minimize human errors and streamline compliance workflows.
- **Focus on Strategic Analysis:**
- Free up resources to focus on forward-looking analyses, such as stress testing and capital planning.

HKMA's paper: Generative AI in Financial Services Space

"Fostering innovation within a well-defined risk management framework is key to the sustainable development of the financial services industry."

- Mr Arthur Yuen, Deputy Chief Executive of the HKMA

Reference: Generative Artificial Intelligence in the Financial Services Space https://brdr.hkma.gov.hk/eng/doc-ldg/docId/getPdf/20241118-4-EN/20241118-4-EN.pdf

How GenAI is being adopted in and impacting the financial services sector

Transforming traditional processes

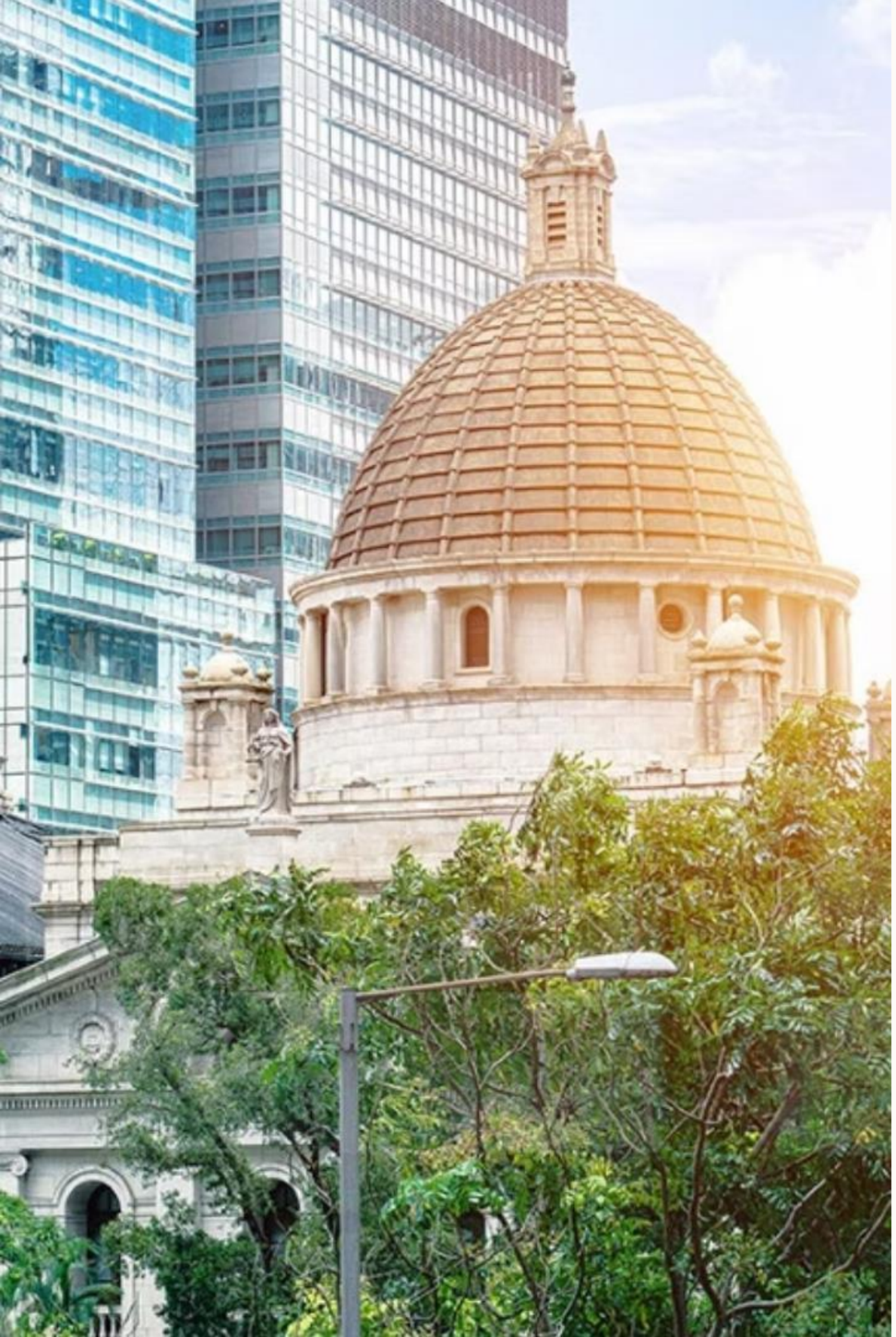
Automating routine tasks, enhancing knowledge management and fraud detection.

Rapid Innovation

Increasingly piloting to improving decision-making, risk management and customer engagement.

Challenges

Potential inaccuracies, algorithmic bias, data privacy issues and cybersecurity risks.

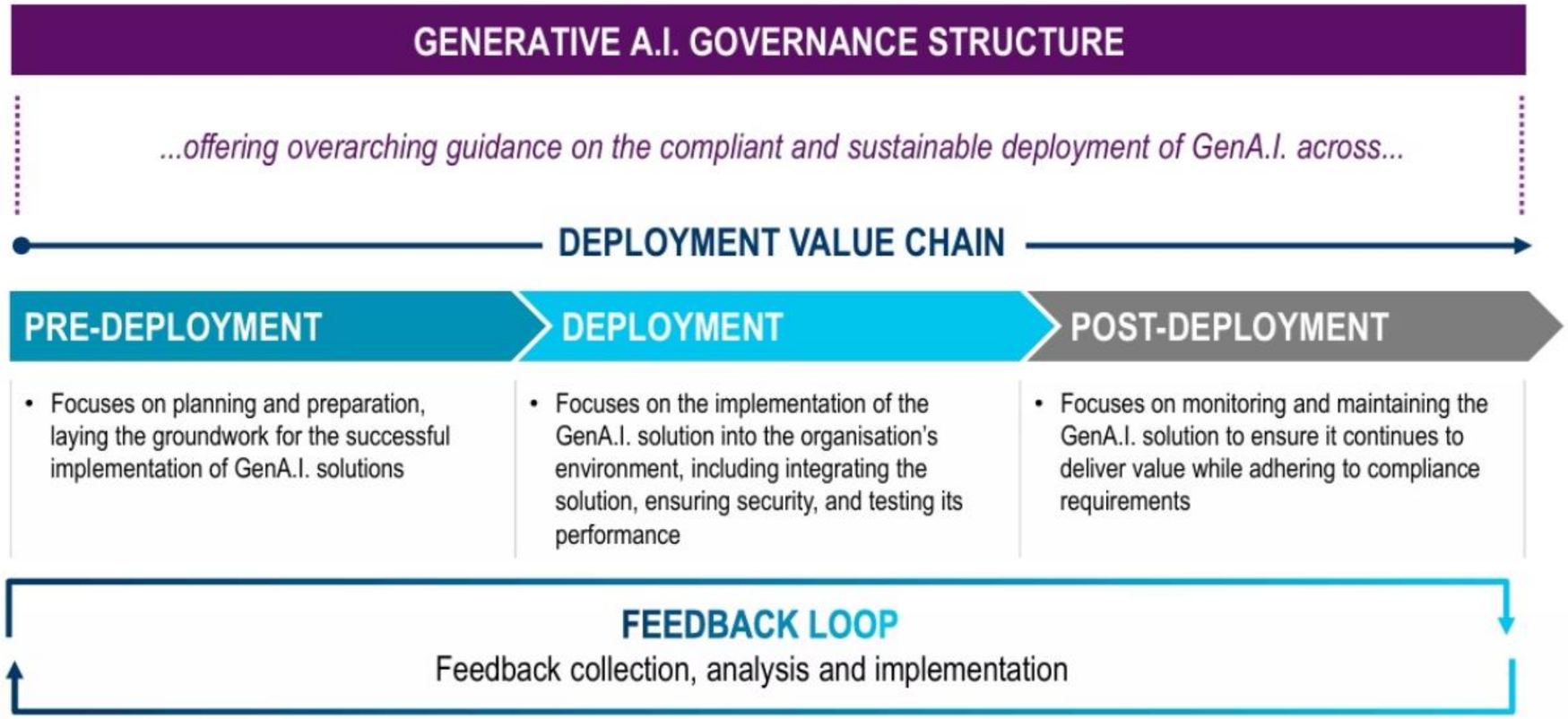


HKMA's initiatives for innovation

- Develop framework and issue guidelines (*Ethical Artificial Intelligence Framework (2024)*)
- Establish **Gen A.I. Sandbox**:
  - Allows institutions to test innovative use cases and gather valuable supervisory feedback from HKMA
  - **Benefit:** identify risks, refine use cases and establish best practices before widespread deployment through sandbox experience

Adoption Considerations and Suggested Practices

The paper provides guidance on how to effectively plan, test, deploy and integrate AI solutions in business operations.



Source: Quinlan & Associates analysis

Adoption Considerations and Suggested Practices (cont'd)

Regulatory Principles for Generative AI Governance

Many jurisdictions have provided supervisory guidance based on a set of principles, typically organized into two categories, to ensure the ethical and responsible use of AI

COMMON PRINCIPLES

Governance & Accountability

Assign clear stakeholder responsibilities in the governance structure and ensure an appropriate level of explainability

Fairness

Ensure equal and inclusive treatment of all users, prevent bias, and avoid discrimination against any certain group

Data Privacy & Protection

Implement robust security measures on the system, seek data privacy consent, and communicate the intended usage of data

Transparency & Disclosure

Provide users with clear and sufficient disclosures about the use of A.I. in the service provided to protect their rights

ADDITIONAL PRINCIPLES

Reliability

Operate the A.I. system with resilience, using high-quality data to deliver accurate and reliable responses

Sustainability

Actively monitor and manage A.I.'s environmental and social impacts to ensure sustainable development

Source: Regulatory disclosures, Quinlan & Associates analysis

Adoption Considerations and Suggested Practices (cont'd)

Governance Across the Value Chain

Based on robust governance framework from the outset, organisations can start their GenAI deployment process across the value chain

GENERATIVE A.I. GOVERNANCE STRUCTURE

PRE-DEPLOYMENT

- **Business Case Development**
- **Proof-of-Concept**
  - User Experience / User Interface
  - Model Design
  - Capability Acquisition Approach
  - Business Risk Assessment
  - Business Contingency Planning
  - Resource Requirement
  - Deployment Timeline

DEPLOYMENT

- **System Integration & Data Migration**
  - Application Layer
  - Integration Layer
  - Storage Layer
  - Ingestion Layer
  - Data Source Layer
  - Security Layer
- **Technology Risk Assessment**
- **Technology Contingency Planning**
- **Solution Performance Testing**

POST-DEPLOYMENT

- **Ongoing Monitoring**
  - Output Management
  - Solution Performance Monitoring
  - Compliance Monitoring
- **Training on GenA.I.**
- **Marketing to the Public**

Source: Quinlan & Associates analysis

Conclusion and Reflection

- **Data Preparation is Foundational:** Data cleansing and feature engineering—constituting roughly 60% of my Python project—form the essential groundwork for robust AI models.
- **Flexibility in Model Training:** There’s no universal rule for model training. What matters is rigorously justifying each methodological choice to ensure clarity and credibility.
- **Balancing Innovation with Caution:** While Generative AI offers remarkable gains in innovation and efficiency, its rapid adoption introduces challenges that necessitate balanced risk management.
- **AI as a Strategic Tool:** AI should be leveraged to fulfill specific business needs—not as an end goal. Aligning technology with process requirements remains paramount.

Let’s embrace AI innovation, but with wisdom and responsibility.

Thank you.